YEAR 10 SCIENCE (ADJUSTED)
PHYSICAL SCIENCE
TEST 2

Name: ______________________________ **Mark:** $\frac{\quad}{28}$

Part 1: Multiple Choice

Write your choice for each question in the table below.

MULTIPLE CHOICE ANSWERS	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

1. The crumple zone of a car is designed to:

   (a) allow the vehicle and passengers to stop more slowly.
   (b) decrease the time before the car comes to a stop after a collision.
   (c) allow the chassis to bend.
   (d) create a reaction force that equals the action force.

2. A car manufacturer wants to produce cars that accelerate faster. Which changes in their cars would result in greater acceleration?

   (a) Increase both the forward force by the engine and the mass of the car.
   (b) Decrease both the forward force and the mass of the car.
   (c) Increase the forward force and decrease the mass of the car.
   (d) Decrease the forward force and increase the mass of the car.

3. The rate of change of speed of an object is known as:

   (a) average speed.
   (b) average velocity.
   (c) change in velocity.
   (d) acceleration.

4. The tendency of an object to resist changes in its motion is called:

   (a) inertia.
   (b) weight.
   (c) friction.
   (d) work.

5. Newton's law, known as the law of inertia, is:

   (a) his first law.
   (b) his second law.
   (c) his third law.
   (d) none of the above.

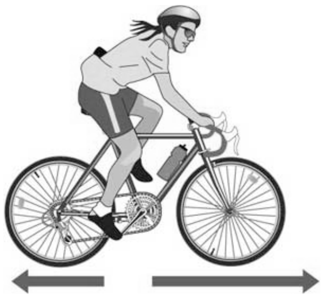
6. An object is acted upon by a force of ***50 N pushing it forwards*** and a total frictional force of ***40 N acting in the opposite direction***. The nett force will be:

   (a) 10 N backwards.
   (b) 10 N forwards.
   (c) 90 N backwards.
   (d) 90 N forwards.

7. The force of gravity acting on an object is called:

   (a) inertia.
   (b) weight.
   (c) friction.
   (d) work.

8. In the diagram below, the arrows show the size and direction of two force arrows acting on the bike.



   Which of the following describes the effect of the pair of forces on the bike?

   (a) The bike will accelerate forward.
   (b) The bike will change its direction of motion.
   (c) The bike will remain stationary.
   (d) The bike will decelerate.

9. A class is studying Newton's three laws. The teacher asks a student wearing rollerblades to face the wall and push against it. The student moves backwards away from the wall. Which is the best explanation for her movement?

   (a) Her action of pushing resulted in a reaction force by the wall, in the opposite direction to her push.
   (b) She used her feet to push herself away from the wall.
   (c) The push is a force and causes her to accelerate.
   (d) The inertia of the student is responsible for her acceleration.

10. Which of the following statements is ***correct***?

   (a) A car at rest has at least two forces acting on it.
   (b) A car at rest has no forces acting on it.
   (c) When you take your foot off the car's accelerator, the car slows down because there are no more forces acting on it.
   (d) When you brake in a car that is rolling down a hill, the car stops because all of the forces add up to zero.

Part 2: Short Answer

1. Explain why it is dangerous to have large parcels or sharp objects loose in the back of a moving car when it brakes sharply to a stop or has an accident. (2 marks)

2. A car moving at 5.0 m/s accelerates at 3.50 m/s^2 for 8.0 s on a straight road. What is the final speed of the car? (***Set your working out clearly.***) (3 marks)
(Hint: Use $v = u + at$)

3. A boat travelling at 8.0 m/s takes 35 s to come to a stop near a boat ramp. Calculate the ***deceleration*** of the boat. (***Set your working out clearly.***) (3 marks)
(Hint: Use $a = \frac{v - u}{t}$)

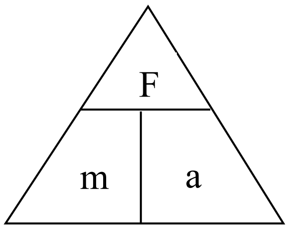
4. Describe the motion of a ball placed on the middle of the back seat of a car in the following situations.

   (a) The car brakes suddenly. ______________________________ (1 mark)

   (b) The car turns right. ______________________________ (1 mark)

   (c) The car accelerates. ______________________________ (1 mark)

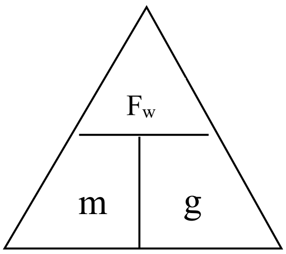
5. Complete the following table on ***forces*** and ***acceleration***. (3 marks)



Force	Mass	Acceleration
	1200 kg	5 m/s^2
150 N	25 kg	
60 N		2.5 m/s^2

6. An object has a mass of 6.0 kg.

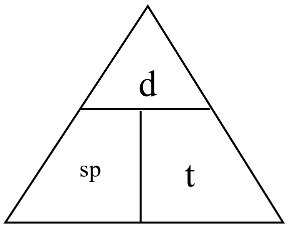
(a) What is its ***weight*** on Earth, if the acceleration due to gravity is 9.8 m/s^2? (2 marks)



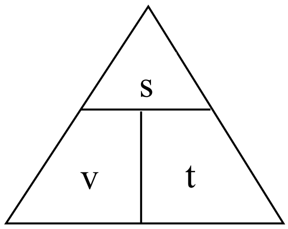
(b) Would the object ***weigh more or less*** on the Moon? Explain your answer. (2 marks)

YEAR 10 SCIENCE
PHYSICAL SCIENCES
PHYSICS DATA SHEET

Speed speed $= \frac{d}{t}$

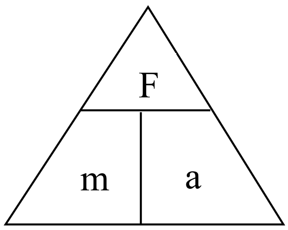


Velocity $v = \frac{s}{t}$

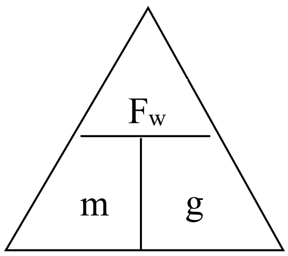


Acceleration $a = \frac{v - u}{t}$ $v = u + at$

Force $F = ma$



Weight $F_w = mg$



Acceleration due to gravity $g = 9.80\ ms^{-2}$

Converting km/h to m/s ***Divide*** the speed in km/h by 3.6 to get the speed in m/s.